CO-2 - AT-3

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Course : Computer Vision

Code : ITA0502

A photograph captured in dimly can be enhanced using gray level transformations

Steps:

- Read iput
- Convert image into gray scale
- pixel values btw 0 to 1
- Apply gamma transformations
- Display enhanced image.

2) Histogram equalization improves image contrast by redistributing pixel intensity across available rang

Steps:

- Read image
- Convert grayscale & calculate image histogram
- Compute CDF & map original intensity
- Display enhanced image

3) Salt & pepper noise appears as random white & black pixels, A median filter can remove these

Steps:

- Read noisy image & select suitable kernel as 3x3
- Replace each pixel with median of it's neighbouring pixels
- Display the filtered image

4) The Laplacian operator detects rapid intensity changes in an image. It enhances the edges of copper tracks

Steps:

- Read image
- Convert to gray scale & calculate Laplacian of image
- Subtract Laplacian from original image
- Display the sharpened result.

5) A low-pass filter reduces high-frequency components associated with fine details & noise

Steps:

- Read the image & convert to gray scale
- Apply fourier transfor & create gaussian low-pass filter
- Multiply the fourier spectrum by filter
- Apply inverse fourier transform
- Display the smoothed image.

6) A high pass filter preserves high frequency component such as edges & fine structures. It can enhance thin blood vessels

Steps:
- Read the retinal image & convert to grayscale
- Apply fourier transform & create high-pass filter
- Multiply frequency by filter & apply inverse fourier
- Display the enhanced image

7) Point detection identifies isolated pixels that differ significantly from their surrounding.

Steps:
- Convert bottle image to gray scale
- Apply point detection using mask
- Calculate absolute filter response
- Display detected points & lines.

8) Adaptive thresholding seperates handwriting from the paper background

Steps:
- Read the answer sheet & convert to grayscale
- Apply gaussian adaptive thresholding
- Using binary inversion so that dark handwriting becomes white
- Display segmented image

9) Region growing is an image segmentation that starts from a seed point & satisfy similarity conditions

Steps:
- Read satellite & convert it to grayscale
- Compare pixel intensities & include pixels that satisfy similarity threshold
- Continue until no pixels satisfying
- Display segmented lake.

10) Edge linking connects broken edge segments into continuous structures. Boundary detection identifies

Steps:
- Read rear image & convert to gray scale
- Apply gaussian blur to reduce noise
- Use canny edge detector
- Use hough transform
- Find relevant edge segments
- Display the detected outline